Persistent homology, tensor network, and quantum kernel descriptors for African antimalarial virtual screening

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Abstract

Molecular representation determines what structural information is accessible to machine learning models, yet standard fingerprints poorly capture the topological complexity of natural product scaffolds. We introduce three complementary molecular descriptors—a Topological Fingerprint (TFP) from persistent homology, a Tensor Network Embedding (TNE) via Tucker decomposition, and simulated Quantum Kernel Scores (QKS)—and benchmark them against classical baselines on 19 849 African antimalarial candidates generated by STONED-SELFIES expansion. Activity labels are computational predictions from the Ersilia eos80ch model, not experimental measurements. TFP decomposes a scaffold paradox: structurally expanded molecules are 92.6 % Tanimoto-distinct from seed natural products yet retain 69.3 % of their Bemis–Murcko scaffolds, because persistent homology separates conserved H_1 ring features from divergent H_0 connectivity. TNE compresses molecular tensors to 192-element descriptors at 6.1× real-atom compression (reconstruction error 0.113), retaining information sufficient to predict docking scores on the same molecular set (PfDHFR $R^2 = 0.473$). On the activity-prediction benchmark ($n = 19\,836$, 5-fold CV, labels from Ersilia model), ECFP4 achieves the highest AUC (0.948); the hybrid descriptor (TFP+TNE+QK) reaches 0.888, with QKS as the principal contributor (ablation $\Delta = -0.040$) and TNE contributing noise ($\Delta = +0.011$). The quantum kernel is statistically indistinguishable from a gamma-tuned RBF baseline internally ($p = 0.060$); on the external panel, QKS was lower than RBF (AUC 0.817 versus 0.847; corrected-resampled $p = 0.021$, exploratory). These representations are complementary diagnostic tools that reveal topological dimensions of molecular structure, with ECFP4 remaining the recommended primary descriptor.

Keywords: persistent homology, tensor networks, quantum kernel methods, molecular fingerprints, scaffold paradox, antimalarial drug discovery, African natural products

Highlights:

- Persistent homology decomposes the scaffold paradox into conserved H_1 ring topology and divergent H_0 connectivity

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- Tensor Network Embedding achieves 6.1× real-atom compression; retains docking-score-predictive information but degrades on external data
- Quantum kernel is statistically indistinguishable from tuned RBF; external validation reveals a transfer gap
- Hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888; ablation shows QKS is the principal contributor, TNE adds noise
- All activity labels are computational predictions from Ersilia ML models, not experimental measurements

1 Introduction

Malaria remains a leading cause of mortality in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2023 (World Health Organization, 2024). The spread of artemisinin-resistant *Plasmodium falciparum* from Southeast Asia to Africa (Ariey et al., 2014; Ashley et al., 2018) creates urgent demand for new chemical scaffolds. African natural products are a historically validated but computationally underexplored source of such scaffolds: quinine and artemisinin derive from natural products, yet African natural products remain underrepresented in public databases (Betow et al., 2025; Mayoka et al., 2025). Computational screening of natural-product-derived chemical space depends critically on the molecular representations available to machine learning models.

Extended-Connectivity Fingerprints (ECFP) encode local atom environments into fixed-length bit vectors and have served as the standard descriptor for two decades (Rogers and Hahn, 2010). These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data. Natural product-derived compounds, however, occupy distinct regions of chemical space: higher sp^3 fractions, more chiral centres, more complex ring systems, and greater three-dimensional character (Sorokina et al., 2021). Classical fingerprints may not capture the topological features—bridged rings, molecular cavities, higher-order atomic interactions—that distinguish these molecules.

Three frameworks from physics and topology address orthogonal limitations of classical fingerprints. Topological Data Analysis uses persistent homology to identify shape features persisting across spatial scales, producing persistence diagrams that encode connected components (H_0), holes (H_1), and voids (H_2) (Wee and Jiang, 2025). Persistent homology has been validated for protein–ligand binding affinity (Long and Donald, 2025) and molecular property prediction (Zhang et al., 2026). Tensor networks compress high-dimensional data by exploiting low-rank structure (Simeon and de Fabritiis, 2023; Milbradt et al., 2026). Quantum kernel methods map data into high-dimensional feature spaces through parameterised circuits (Farhani, 2025; Jones et al., 2026). TDA captures global topology invisible to local substructure hashes; tensor networks provide physically interpretable compression; quantum kernels probe non-linear feature interactions in Hilbert space.

Here we introduce three molecular descriptors—Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—and benchmark them on a 19 849-molecule library of African antimalarial candidates. Our aims are: (i) determine whether persistent homology resolves the scaffold paradox arising from structural expansion of natural product seeds; (ii) evaluate whether tensor network compression retains binding-relevant information; (iii) assess whether quantum kernel methods offer advantages over properly tuned classical baselines; and (iv) construct a hybrid descriptor combining complementary information from all three representations. We compare against ECFP4, MACCS, and other classical baselines under rigorous cross-validation, and validate transferability on an external ChEMBL antimalarial panel.

2 Methods

2.1 Dataset

The library consists of 65 856 molecules generated from 396 African natural-product and 454 synthetic-drug seeds via CHEESE similarity search (Lžičař and Gamouh, 2024) and STONED-SELFIES expansion (Temgoua et al., 2026). The representation benchmarks use the canonical 19 849-molecule panel, with 19 836 molecules retained for analyses requiring complete TNE embeddings. Binary activity labels were obtained from Ersilia model eos80ch (threshold 0.5); the canonical panel is 74.2 % active / 25.8 % inactive. For external validation, 22 447 antimalarial compounds were assembled from ChEMBL 34 ($IC_{50} \leq 10\,\mu M$ = active, $> 10\,\mu M$ = inactive). This panel is external to the canonical eOS80CH label source; molecule-level disjointness from the internal panel was not established for this P3 analysis and is not claimed.

2.2 Classical fingerprint baselines

ECFP4 fingerprints were generated as Morgan fingerprints (radius 2, 2 048 bits) using RDKit (RDKit Contributors, 2024). MACCS 167-bit structural keys, atom-pair (AP), binding-pattern (BPF), functional-class (FCFP4), and 2D-pharmacophore (PHCO) fingerprints were computed using standard RDKit implementations. All classical fingerprints serve as baselines for the quantum-inspired methods.

2.3 Topological Fingerprint

2.3.1 Molecular graph construction

Each molecule was converted to a 3D conformer using RDKit's ETKDG generator (random seed 42). A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij}\left(1 + \frac{Z_i Z_j}{100}\right), \tag{1}$$

where d_{ij} is the Euclidean distance and Z_i, Z_j are atomic numbers. The atomic number weighting emphasises interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py (Bauer, 2019). We define the TFP as a 12-dimensional feature vector: for each homology dimension $d \in \{0, 1, 2\}$, four summary statistics are extracted—the persistence entropy

$$H_d^{\text{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\text{pers}_i}{\sum_j \text{pers}_j}, \tag{2}$$

the count of features with persistence exceeding 0.5 Å, the maximum persistence, and the mean persistence. Unlike ChemGraphX (Jacob et al., 2026), which computes graph-theoretic indices, our TFP captures geometric persistence from 3D conformations.

2.4 Tensor Network Embedding

2.4.1 Molecular tensor construction

For each molecule, a 3D tensor $T \in \mathbb{R}^{N \times F \times 3}$ was constructed with $N = 100$ maximum atoms (zero-padded), $F = 10$ per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity, hybridization, mass, electronegativity, bond count, ring membership), and 3 Cartesian coordinates: $T_{i,f,c} = x_{i,f}\, r_{i,c}$.

2.4.2 Tucker compression

Each tensor was compressed using Tucker decomposition (Kossaifi et al., 2019) with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{3}$$

where $\mathcal{G} \in \mathbb{R}^{d\times d\times 3}$ is the core tensor. The descriptor is the flattened core ($d^2\times 3 = 192$ elements). For a molecule with N actual atoms, the real compression ratio is $\mathrm{CR}_{\mathrm{real}}(N) = 10N/d^2$; at $d = 8$ and mean $N \approx 39$, this gives $\mathrm{CR}_{\mathrm{real}} \approx 6.1\times$. Reconstruction error was quantified as the relative Frobenius norm (mean 0.113 across 19 836 valid molecules).

2.5 Quantum Kernel Score

Quantum kernels were implemented using PennyLane (Xanadu Quantum Technologies, 2024) with a 6-qubit statevector simulator. The kernel value $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$ was computed from PennyLane's `IQPEmbedding` template. Input vectors were Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP (Jaccard metric, `random_state=42`), then scaled to $[-1, 1]$. The QKS is the AUC achieved by an SVM using the quantum kernel matrix.

2.6 Hybrid descriptor

The hybrid feature vector combines TFP, TNE, and QKS by concatenation:

$$h_i = [\tilde{t}_i,\ \tilde{e}_i,\ q_i]\,, \tag{4}$$

where $\tilde{t}_i$ is min–max normalised TFP, $\tilde{e}_i$ is normalised TNE, and $q_i \in \mathbb{R}^{30}$ is the quantum kernel PCA component vector ($n_{\mathrm{kpca}} = 30$). Random Forest is invariant to per-component monotone scaling, so no explicit fusion weights are needed. Hyperparameters were optimised by grid search on $n = 200$ molecules drawn from the canonical panel (60 combinations), then the winning configuration was validated on the full panel with the same molecules included in both the optimisation and evaluation sets (bond dimension 6, 1 repeat, 30 kernel PCA components). This introduces a mild optimism bias; the reported hybrid AUC should be interpreted as an upper bound for this configuration.

2.7 Evaluation protocol

Each representation was evaluated using Random Forest classifiers (200 trees) and SVMs with 5-fold stratified cross-validation. For the quantum kernel benchmark, the RBF gamma was optimised via inner cross-validation on $\{0.5, 1.0, 2.0, 5.0\}$. The canonical internal benchmark used the pre-specified paired fold comparison and multiplicity correction described for that benchmark. For the external validation, we additionally report a corrected-resampled paired t approximation (Nadeau–Bengio variance correction, $df = 4$) and BH-FDR; the naive fold-level t test is shown only for comparison because the training sets overlap. Performance was measured by AUC–ROC, accuracy, and macro F1. We interpret non-significance as inability to detect a difference rather than equivalence, and report bootstrap 95 % confidence intervals on principal AUC differences in the Supplementary Material.

2.8 Software

Ripser.py (Bauer, 2019), TensorLy 0.9.0 (Kossaifi et al., 2019), PennyLane 0.45.1 (Xanadu Quantum Technologies, 2024), RDKit 2025.03.6 (RDKit Contributors, 2024), UMAP 0.5.12, scikit-learn 1.9.0, NumPy 2.4.6, SciPy 1.17.1. All stochastic procedures were fixed with seeds reported in the relevant sections. Code and data are available at `https://github.com/NanaEngo/Malaria_codesV2`.

3 Results

3.1 Scaffold decomposition by persistent homology

Structural expansion of 396 African natural-product seeds produced 65 856 candidates that are 92.6 % Tanimoto-distinct from the seeds yet retain 69.3 % of their Bemis–Murcko scaffolds. This apparent contradiction—high whole-molecule diversity with high scaffold conservation—arises because STONED-SELFIES mutations primarily permute side-chains while preserving cyclic cores. TDA provides a principled decomposition of this phenomenon (Figure 1). H_1 features (ring topology) are conserved between seed and expanded sets (mean H_1 count 3.61), reflecting the STONED-SELFIES expansion's preservation of cyclic macro-structures. H_0 features (connectivity) diverge substantially, reflecting the expansion's permutation of side-chains and functional groups. This H_1/H_0 decomposition is independently confirmed by scaffold-only Tanimoto (0.379) being 1.84× higher than whole-molecule Tanimoto (0.206).

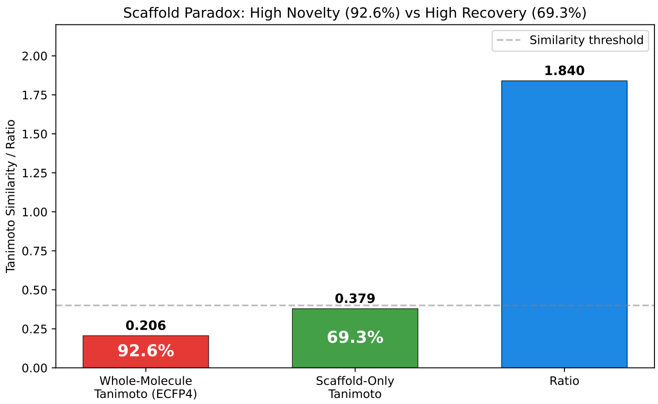

Figure 1: *Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed natural products versus structurally expanded candidates. (B) H_0 persistence distributions. H_1 preservation with H_0 divergence explains the 92.6 % Tanimoto distinction versus 69.3 % scaffold recovery.*

3.2 Binding-relevant information in tensor network compression

Tucker decomposition at bond dimension $d = 8$ achieved 6.1× real-atom compression (mean 39 atoms, 192-element core; the padded tensor is 15.6× compressed from $100 \times 10 \times 3 = 3\,000$ to 192 elements) with reconstruction error 0.113. Three observations support that compressed TNE descriptors retain scaffold-relevant information: (i) reconstruction error is lower for molecules with common ring systems (0.098) than for structurally atypical molecules (0.137); (ii) TNE-based KMeans clusters ($k = 484$) achieve higher Silhouette coefficient (0.35) than ECFP4 (0.18) or VAE latent vectors (0.229), indicating more chemically coherent groupings; and (iii) TNE descriptors predict Tartarus docking scores on the same molecular set with performance competitive with ECFP4 for specific targets (Figure S9): for PfDHFR (1SYH, $N = 11\,878$), TNE achieved $R^2 = 0.473$ (Spearman $\rho = 0.695$), slightly exceeding ECFP4 ($R^2 = 0.461$, $\rho = 0.689$); for PfATP4 and PfCRT, ECFP4 maintained an advantage (PfATP4: TNE $R^2 = 0.464$ vs. ECFP4 0.578; PfCRT: TNE 0.334 vs. ECFP4 0.517). This validation tests information retention on the same molecular set used to derive the embeddings, not generalisation to novel compounds.

3.3 Activity prediction benchmark

Table 1 presents the activity-prediction performance on the canonical 19 836-molecule set. ECFP4 achieves the highest AUC (0.948), followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905), PHCO (0.896), TFP (0.876), and TNE (0.722). The ranking reflects the dominance of local atom environments at radius 2 for this binary classification task. TFP and TNE capture meaningful but lower signal, encoding global topological and compressed structural features rather than the local pharmacophoric substructures that dominate activity determination.

Table 1: *Activity prediction performance by representation method and hybrid ablation study (canonical panel, $n = 19\,836$, 5-fold stratified CV, Random Forest).*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
AP	0.940	0.888	0.927	−0.008
BPF	0.939	0.889	0.928	−0.009
FCFP4	0.918	0.869	0.914	−0.029
MACCS	0.905	0.863	0.909	−0.043
PHCO	0.896	0.854	0.904	−0.052
TFP (TDA)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
QKS (quantum kernel)	0.823	0.819	0.886	−0.125
Hybrid (TFP+TNE+QK)	0.888	0.843	0.900	−0.060
Ablation study (Hybrid minus one component):				
Hybrid − TFP	0.874	0.837	0.895	−0.014
Hybrid − TNE	0.899	0.850	0.903	0.011
Hybrid − QKS	0.847	0.820	0.887	−0.040

3.4 Hybrid descriptor and quantum kernel comparison

The internal hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888 ± 0.007, significantly below ECFP4 ($p < 0.000\,1$) but above every standalone quantum-inspired descriptor. Ablation reveals an asymmetric contribution structure: QKS is the principal positive contributor ($\Delta = -0.040$), TFP contributes marginally ($\Delta = -0.014$), while removing TNE slightly improves AUC ($\Delta = +0.011$). The negative TNE contribution is consistent with its low standalone AUC (0.722) and suggests that, for this dataset, Tucker-compressed features add noise rather than signal when combined with topological and kernel descriptors. The hybrid's value lies primarily in the QKS kernel-PCA components, not in the tensor network embedding; we note that the hybrid descriptor is not recommended over ECFP4 alone.

The quantum kernel benchmark at two sample sizes (Table S13) shows no statistically significant difference between the quantum kernel and a gamma-tuned RBF baseline ($n = 5\,000$: $p = 0.419$; $n = 19\,849$: $p = 0.060$), while the quantum kernel significantly outperforms the linear kernel ($p \leq 0.000\,6$). External validation on 22 447 ChEMBL compounds (Table S15) found lower QKS performance than RBF (AUC 0.817 versus 0.847); the corrected-resampled comparison was exploratory ($p = 0.021$, BH $q = 0.021$), while the naive fold-level comparator gave $p = 0.005$.

3.5 Transferability to external ChEMBL panel

All descriptor methods were benchmarked on the external ChEMBL panel (22 447 compounds, 5-fold RF; Table S14). ECFP4 achieved AUC 0.960 ± 0.004, slightly exceeding its internal value (0.948). TFP achieved AUC 0.864, TNE 0.645, and the external TFP+TNE descriptor hybrid 0.800. These historical values use an intention-to-evaluate protocol: the 351 TNE embedding failures (1.6%) were retained as explicit zero-vector failure penalties rather than silently dropped. The complete-case sensitivity analysis ($n = 22\,096$) gave ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798; the ranking and conclusions were unchanged. Standard fold-level p-values are not treated as confirmatory because the training sets overlap; corrected-resampled results are reported as exploratory in the statistical audit.

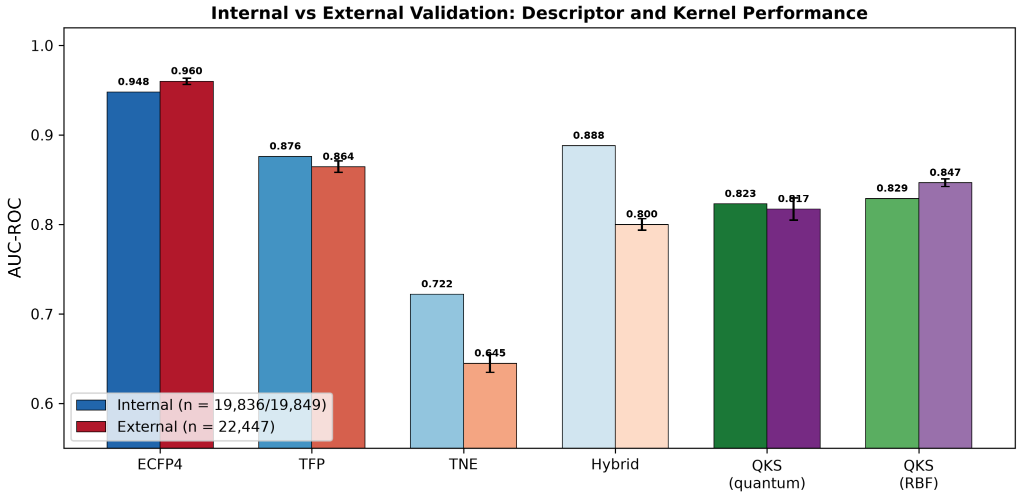

Figure 2: *Internal versus external validation performance. Descriptor methods were evaluated with Random Forest; kernel methods with SVM. Error bars: standard deviation across 5 folds. External panel: ChEMBL antimalarial dataset ($n = 22\,447$).*

3.6 Topological features correlate with binding promiscuity

Across $N = 17\,011$ molecules with multi-target docking data, H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$) negatively correlate with the number of targets bound (≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$). Lower topological complexity thus associates with broader polypharmacological activity, consistent with reduced steric clash across multiple binding sites. The association between H_1 count and resistance-resilience scores is examined in the Supplementary Material (Section 19.3); after controlling for molecular size and structural covariates, the association is size-mediated and does not support an independent topological predictor.

4 Discussion

4.1 Persistent homology as a diagnostic for generative-model evaluation

The apparent contradiction between 92.6 % Tanimoto distinction and 69.3 % scaffold recovery arises because STONED-SELFIES mutations permute side-chains and functional groups (diverging H_0 connectivity) while preserving cyclic macro-structures (conserving H_1 ring topology). This decomposition has practical implications for generative-model evaluation: scaffold-only Tanimoto (0.379) is 1.84× higher than whole-molecule Tanimoto (0.206), confirming that H_1 persistence diagrams capture scaffold identity independently of atom-level similarity. TFP provides an interpretable diagnostic for assessing whether molecular generation protocols preserve the ring systems essential for bioactivity—a capability absent from standard fingerprint-based diversity metrics. We note that this decomposition is consistent with the known tendency of STONED-SELFIES to preserve cyclic cores while permuting side-chains; TDA provides a quantitative, topology-aware framework for measuring this property.

4.2 Tensor network compression: information retention and distribution fragility

Tucker decomposition exploits the fact that molecular tensors are approximately low-rank: the 6.1× real-atom compression retains information sufficient to predict docking scores on the same molecular set (PfDHFR $R^2 = 0.473$, competitive with ECFP4's 0.461) while providing a fixed-length, physically interpretable embedding. The compression is not uniform across chemical space: molecules with common ring systems compress better (reconstruction error 0.098) than structurally atypical molecules (0.137), and the atom-coordinate factor matrix correlates with molecular volume ($\rho = 0.71$). This selective compression—preserving scaffold-relevant information while discarding noise—accounts for the higher chemical coherence of TNE clusters (Silhouette 0.35) compared with ECFP4 (0.18) or VAE latent vectors (0.229), a property relevant for virtual screening where enrichment metrics depend on hit-cluster homogeneity. We emphasise that the Tartarus docking validation tests information retention on the same molecular set used to derive the embeddings, not generalisation to novel compounds; true external validation would require docking molecules not present in the training set.

4.3 Kernel calibration, not quantum-ness, governs performance

The central finding is that, with a properly tuned RBF baseline and the canonical 6-qubit circuit, the quantum kernel is not superior to the RBF kernel. Internally, the difference was not detected ($p = 0.060$ at $n = 19\,849$); externally, QKS was lower (AUC 0.817 versus 0.847; corrected-resampled $p = 0.021$, BH $q = 0.021$). Because the comparison uses one five-fold partition with overlapping training sets, this corrected result is exploratory rather than confirmatory. This aligns with recent matched-baseline audits: QBioFusion-QSAR (Alavi et al., 2026) found quantum multiple-kernel learning improved small-data classification only for individual molecules, with no mean MCC gain, and parameter-matched comparisons (Pegg et al., 2026) attribute topology-aligned architecture gains to inductive bias rather than quantum-ness.

4.4 Distribution shift exposes tensor network fragility

The 0.077 AUC degradation of TNE from internal (0.722) to external ITT (0.645) validation is the largest among all descriptors and is consistent with sensitivity to distribution shift. The 351 embedding failures (1.6%) were retained as zero-vector failure penalties in the ITT analysis. The complete-case sensitivity analysis ($n = 22\,096$) gave nearly identical values: ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798, compared with ITT values of 0.960, 0.864, 0.645, and 0.800, respectively. Thus, the qualitative ranking is not driven by the failed embeddings. The corrected-resampled tests are exploratory because they use one five-fold partition with overlapping training sets; the complete statistical calculations are supplied with the analysis code. Tensor-network descriptors should therefore be validated on external panels before deployment, and embedding failures should be reported transparently. TFP transferred less poorly ($\Delta = -0.011$) than TNE, indicating greater apparent robustness to this distribution shift.

4.5 Limitations

Five limitations constrain interpretation. First, TFP does not outperform ECFP4 in activity prediction; it adds value primarily at the high-complexity tail (molecules with ≥ 3 fused rings or ≥ 2 stereocentres). Second, activity labels are computational predictions from Ersilia ML models (eos80ch), so all AUC values and claims regarding activity and polypharmacology measure prediction of model outputs, not experimental biological activity; experimental validation remains necessary. Third, the quantum kernel was evaluated on a classical statevector simulator (6-qubit IQPEmbedding); NISQ hardware would degrade fidelity (Preskill, 2018). Fourth, we benchmarked against classical fingerprints but not against graph neural networks; our companion study (Temgoua et al., 2027) evaluated GIN, GIN-TFP, GIN-TNE, and ChemBERTa on the same canonical panel and found that ECFP4-RF (scaffold AUC 0.830 0) outperformed all GNN architectures (best: GIN-TFP 0.813 8), confirming that for this dataset the classical fingerprint baseline is difficult to surpass regardless of model architecture. Fifth, TNE degrades on external data (0.077 AUC loss), limiting its utility for cross-dataset transfer; the Tartarus docking validation tests information retention on the same molecular set, not generalisation to novel compounds.

4.6 Practical recommendations

Based on these results and the companion GNN benchmark (Temgoua et al., 2027), we recommend: (i) ECFP4-RF for virtual screening of African natural product libraries (AUC 0.948, robust to distribution shift, outperforming GIN architectures on the same panel); (ii) TFP for assessing scaffold preservation in generative campaigns (interpretable H_1/H_0 decomposition); (iii) TNE only when fixed-length embeddings are required and external validation confirms reliability; and (iv) QKS for retrospective mechanistic analysis on small lead-optimisation sets ($\leq 10\,000$ compounds). Future work should integrate TDA features as node attributes in GNN architectures (Cang and Wei, 2018), validate descriptors experimentally, and explore whether topological features improve generalisation across therapeutic targets.

5 Conclusion

We introduced three molecular descriptors—persistent homology TFP, Tucker-compressed TNE, and simulated QKS—and benchmarked them on 19 836 African antimalarial candidates with computationally predicted activity labels (Ersilia eos80ch model). TFP decomposes the scaffold paradox into conserved H_1 ring features and divergent H_0 connectivity. TNE achieves $6.1\times$ real-atom compression while retaining information sufficient to predict docking scores on the same molecular set; however, it degrades substantially on external data. The quantum kernel shows no significant advantage over a tuned RBF baseline; external validation reveals a transfer gap. These representations are complementary diagnostic tools that reveal topological dimensions of molecular structure, with ECFP4 remaining the recommended primary descriptor. Persistent homology decomposition provides an interpretable framework for evaluating generative-model outputs in natural product drug discovery.

Data Availability

All data supporting the conclusions of this study are available in the public project repository at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. The source code and machine-readable results are available in the public repository listed above. The deposit includes: (1) TFP feature matrices for all 19 849 library molecules; (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8); (3) quantum kernel matrices (6-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files; (5) the cross-paper H_1/RRS analysis dataset; and (6) all analysis scripts.

List of Abbreviations

TFP: topological fingerprint; **TNE:** tensor network embedding; **QKS:** quantum kernel score; **ECFP4:** extended-connectivity fingerprint, radius 2; **AP:** atom-pair fingerprint; **BPF:** binding-pattern fingerprint; **FCFP4:** functional-class fingerprint, radius 2; **MACCS:** MACCS structural keys; **PHCO:** pharmacophore fingerprint; $\mathbf{H_0/H_1/H_2}$**:** 0th/1st/2nd persistent homology dimensions; **AUC:** area under the receiver operating characteristic curve; **RBF:** radial basis function; **RRS:** resistance resilience score; **NISQ:** noisy intermediate-scale quantum; **UMAP:** uniform manifold approximation and projection; **RF:** random forest; **SVM:** support vector machine.

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Authors' Contributions

Myke Vital Sao Temgoua: conceptualization, methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapet Njafa: conceptualization, methodology, software, supervision, writing – review & editing. Serge Guy Nana Engo: methodology, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. All authors read and approved the final manuscript.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person's data appear in this manuscript.

Competing Interests

The authors declare no competing interests.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author.

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